

Chapter 13

Mathematical Models

Why Should We Study Mathematical Models?

Mathematical models help us understand and predict real-world systems. They are widely used in:

- Agriculture
- Economics
- Engineering
- Biology
- Environmental Science
- Data Analysis

Models help scientists and engineers make better decisions using mathematics.

13.1 Introduction

A mathematical model is a mathematical representation of a real-life system. It converts practical problems into mathematical equations.

Models help us:

- Analyze systems
- Predict future behavior
- Optimize production
- Study growth and decay
- Understand experimental data

13.2 Agricultural Systems

Agricultural systems involve processes related to farming and crop production.

Mathematical models are used in agriculture to study:

- Crop growth

- Irrigation systems
- Fertilizer usage
- Population of pests
- Weather effects
- Yield prediction

Example

Suppose crop yield depends on fertilizer quantity.

If:

$$Y = 5 + 2x$$

where:

- Y = crop yield
- x = fertilizer amount

then the model predicts that increasing fertilizer increases crop production.

13.3 Mathematical Models

Definition of Mathematical Model

A mathematical model is a mathematical equation or relation used to describe a real-life phenomenon.

13.3.1 Characteristics of a Good Model

- Simple and understandable
- Accurate
- Predictive
- Based on real data
- Easy to analyze

13.4 Classification of Mathematical Models

Mathematical models can be classified into different categories depending on their structure, behavior and applications.

Classification helps us understand which model is suitable for a particular real-life problem.

Why Classification is Important?

Different real-world problems require different types of models.

For example:

- Population growth uses dynamic models.
- Weather prediction uses probabilistic models.
- Cost analysis uses linear models.
- Crop growth may use nonlinear models.

Thus, classification helps in selecting the correct mathematical model.

13.5 Main Types of Mathematical Models

Mathematical models are mainly classified as:

1. Deterministic Models
2. Probabilistic Models
3. Linear Models
4. Nonlinear Models
5. Static Models
6. Dynamic Models
7. Discrete Models
8. Continuous Models

13.5.1 Deterministic Models

Definition of Deterministic Model

A deterministic model gives a fixed output for a given input.
There is no randomness or uncertainty in the model.

Characteristics

- Predictable results
- No random variables
- Same input gives same output
- Easier to analyze mathematically

Example

$$y = 2x + 5$$

If:

$$x = 3$$

then:

$$y = 11$$

The result will always remain the same.

Applications

- Engineering calculations
- Cost estimation
- Motion equations
- Structural analysis

13.5.2 Probabilistic Models**Definition of Probabilistic Model**

A probabilistic model includes randomness and uncertainty. The output may vary even for the same input.

Characteristics

- Contains random variables
- Output is uncertain
- Based on probability theory
- Used when uncertainty exists

Example

Weather forecasting models.

Rain may or may not occur even under similar conditions.

Applications

- Weather prediction
- Stock market analysis
- Population studies
- Risk analysis

13.5.3 Linear Models

Definition of Linear Model

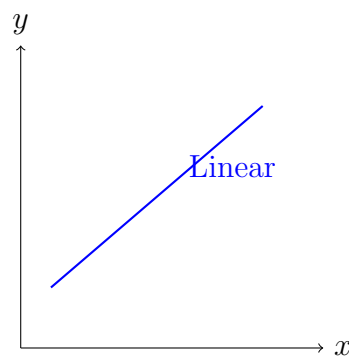
A model is called linear if variables appear with power 1 only.

General Form

$$y = a + bx$$

Characteristics

- Straight-line graph
- Constant rate of change
- Easy calculations
- Simple interpretation



Applications

- Cost analysis
- Income prediction
- Simple growth models

13.5.4 Nonlinear Models

Definition of Nonlinear Model

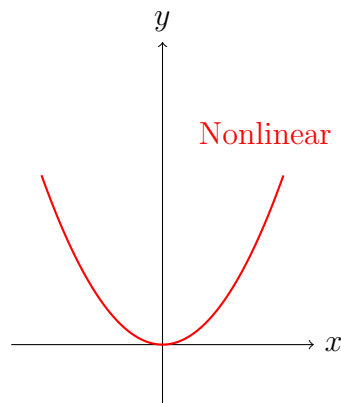
A model is nonlinear if variables have powers greater than 1 or involve trigonometric, exponential or logarithmic functions.

Example

$$y = x^2 + 3x + 1$$

Characteristics

- Curved graphs
- Variable rate of change
- More realistic for many systems
- Difficult calculations



13.5.5 Static Models

Definition of Static Model

A static model does not involve time as a variable.

Example

$$C = 5x + 100$$

where C is cost and x is quantity.

Applications

- Cost analysis
- Market equilibrium
- Structural design

13.5.6 Dynamic Models

Definition of Dynamic Model

A dynamic model changes with time.

Example

Population growth:

$$P(t) = 1000e^{0.02t}$$

Characteristics

- Time-dependent
- Describes changing systems
- Used for prediction

Applications

- Population growth
- Disease spread
- Economic forecasting

13.5.7 Discrete Models

Definition of Discrete Model

A discrete model changes only at specific intervals.

Example

Number of students in a class each year.

Characteristics

- Values are countable
- Time changes step-by-step
- Used in computer simulations

13.5.8 Continuous Models

Definition of Continuous Model

A continuous model changes continuously over time.

Example

Water flow in a river.

Characteristics

- Infinite possible values
- Smooth changes
- Often represented by differential equations

13.6 Summary Table

Model Type	Main Feature
Deterministic	Fixed output
Probabilistic	Random output
Linear	Straight-line relation
Nonlinear	Curved relation
Static	No time dependence
Dynamic	Changes with time
Discrete	Step-by-step changes
Continuous	Continuous changes

Important Notes

- Mathematical models simplify real-world systems.
- Different systems require different models.
- Linear models are simplest.
- Nonlinear models are more realistic.
- Dynamic models involve time.
- Probabilistic models handle uncertainty.

13.7 Practice Questions

1. Define deterministic model.
2. Define probabilistic model.
3. Differentiate between linear and nonlinear models.
4. Explain static and dynamic models.
5. What is a discrete model?

6. What is a continuous model?
7. Give two applications of dynamic models.
8. Why are nonlinear models important?
9. Differentiate between deterministic and probabilistic models.
10. Explain importance of classification of mathematical models.

13.8 Fitting of Linear Model

A linear model has the form:

$$y = a + bx$$

where:

- a = intercept
- b = slope

Experimental Data

x	y
1	3
2	5
3	7
4	9

Observation

As x increases by 1, y increases by 2.

Hence slope:

$$b = 2$$

Using point:

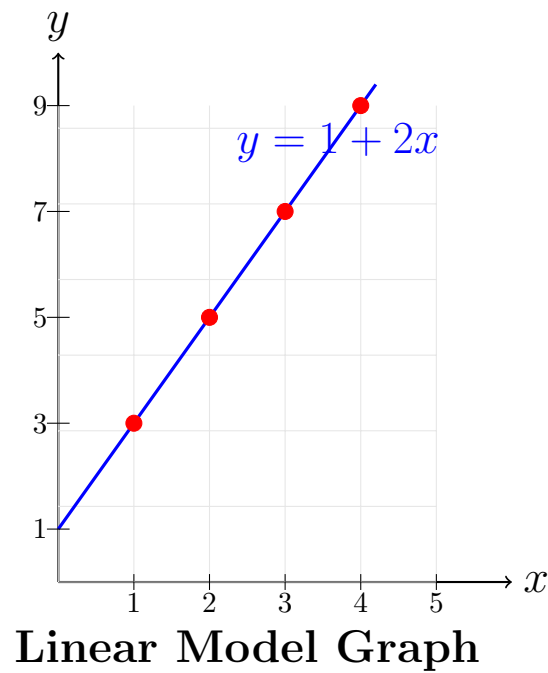
$$(1, 3)$$

$$3 = a + 2(1)$$

$$a = 1$$

Thus linear model is:

$$y = 1 + 2x$$



13.9 Fitting of Quadratic Model

A quadratic model has the form:

$$y = a + bx + cx^2$$

Experimental Data

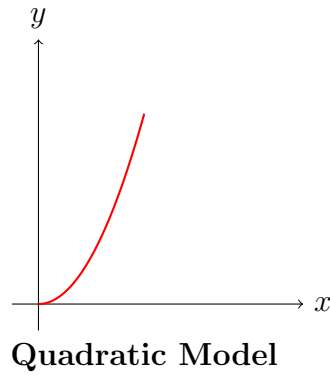
x	y
1	1
2	4
3	9
4	16

Observing the data:

$$y = x^2$$

Hence the quadratic model is:

$$y = x^2$$



13.10 Fitting of Exponential Model

An exponential model has the form:

$$y = ab^x$$

where:

- a = initial value
- b = growth factor

Experimental Data

x	y
0	2
1	4
2	8
3	16

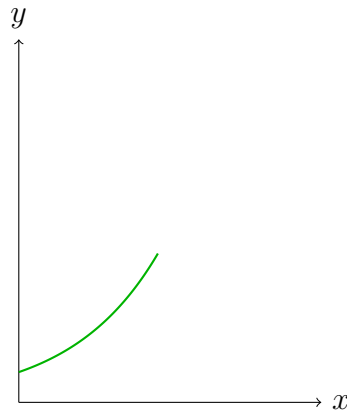
Observe that values double each time.

Hence:

$$a = 2, \quad b = 2$$

Thus model becomes:

$$y = 2(2^x)$$

**Exponential Growth Model**

13.11 Comparison of Models

Model	General Form
Linear	$y=a+bx$
Quadratic	$y=a+bx+cx^2$
Exponential	$y=ab^x$

Important Notes

- Mathematical models simplify real-world systems.
- Linear models represent straight-line relationships.
- Quadratic models represent curved relationships.
- Exponential models represent rapid growth or decay.
- Models help predict future outcomes.
- Experimental data is important for constructing models.

13.12 Solved Examples on Fitting of Models

Solved Example 1 : Linear Model

Fit a linear model for the data:

x	y
1	3
2	5
3	7
4	9

Solution

Linear model:

$$y = a + bx$$

Observe:

y increases by 2

Hence:

$$b = 2$$

Using point:

$$(1, 3)$$

$$3 = a + 2(1)$$

$$a = 1$$

Therefore:

$$y = 1 + 2x$$

Solved Example 2 : Linear Model

Fit a linear model:

x	y
0	2
1	4
2	6
3	8

Solution

Slope:

$$b = 2$$

Using point:

$$(0, 2)$$

$$2 = a + 2(0)$$

$$a = 2$$

Hence:

$$\boxed{y = 2x + 2}$$

Solved Example 3 : Linear Model

Fit a linear model:

x	y
2	7
3	10
4	13
5	16

Solution

Difference:

y increases by 3

Thus:

$$b = 3$$

Using point:

$$(2, 7)$$

$$7 = a + 3(2)$$

$$a = 1$$

Therefore:

$$\boxed{y = 3x + 1}$$

Solved Example 4 : Quadratic Model

Fit a quadratic model:

x	y
1	1
2	4
3	9
4	16

Solution

Observe:

$$y = x^2$$

Hence:

$$y = x^2$$

Solved Example 5 : Quadratic Model

Fit a quadratic model:

x	y
1	3
2	6
3	11
4	18

Solution

Assume:

$$y = a + bx + cx^2$$

Using points:

$$3 = a + b + c$$

$$6 = a + 2b + 4c$$

$$11 = a + 3b + 9c$$

Solving:

$$a = 2, \quad b = 0, \quad c = 1$$

Thus:

$$y = x^2 + 2$$

Solved Example 6 : Quadratic Model

Fit a quadratic model:

x	y
0	1
1	4
2	9
3	16

Solution

Observe:

$$y = x^2 + 2x + 1$$

Checking:

$$x = 0, \quad y = 1$$

$$x = 1, \quad y = 4$$

Correct.

Therefore:

$$y = x^2 + 2x + 1$$

Exponential Model Examples**Solved Example 7 : Exponential Model**

Fit an exponential model:

x	y
0	2
1	4
2	8
3	16

Solution

Exponential form:

$$y = ab^x$$

At:

$$x = 0, \quad y = 2$$

Hence:

$$a = 2$$

Observe values double each time:

$$b = 2$$

Thus:

$$y = 2(2^x)$$

Solved Example 8 : Exponential Model

Fit an exponential model:

x	y
0	3
1	9
2	27
3	81

Solution

Initial value:

$$a = 3$$

Growth factor:

$$b = 3$$

Hence:

$$y = 3(3^x)$$

Solved Example 9 : Exponential Model

Fit an exponential model:

x	y
0	5
1	10
2	20
3	40

Solution

Initial value:

$$a = 5$$

Values double:

$$b = 2$$

Thus:

$$y = 5(2^x)$$

Solved Example 10 : Exponential Model

Fit an exponential model:

x	y
0	1
1	2
2	4
3	8

Solution

Initial value:

$$a = 1$$

Growth factor:

$$b = 2$$

Hence:

$$y = 2^x$$

13.13 Exercise

Unsolved Exercise

1. Fit a linear model:

$$(1, 4), (2, 6), (3, 8)$$

2. Fit a linear model:

$$(0, 5), (1, 8), (2, 11)$$

3. Fit a quadratic model:

$$(1, 1), (2, 4), (3, 9)$$

4. Fit a quadratic model:

$$(0, 1), (1, 4), (2, 9)$$

5. Fit an exponential model:

$$(0, 2), (1, 6), (2, 18)$$

6. Fit an exponential model:

$$(0, 4), (1, 8), (2, 16)$$

7. Identify whether the model is linear or nonlinear:

$$y = 3x + 2$$

8. Identify whether the model is linear or nonlinear:

$$y = x^2 + 1$$

9. Write general form of exponential model.

10. Write general form of quadratic model.

Answer Key

1.

$$y = 2x + 2$$

2.

$$y = 3x + 5$$

3.

$$y = x^2$$

4.

$$y = x^2 + 2x + 1$$

5.

$$y = 2(3^x)$$

6.

$$y = 4(2^x)$$

7. Linear

8. Nonlinear

9.

$$y = ab^x$$

10.

$$y = a + bx + cx^2$$

G. H. Hardy

“A mathematician, like a painter or poet, is a maker of patterns.”

From Hardy's famous book *A Mathematician's Apology*